

Chapter 16 - The Derivatives

1. Derivative and Differentiation
 - a. (Definition/First Principle) of Derivative
2. Rules of Differentiation
3. Parametric Differentiation
4. Implicit Differentiation
5. Successive Differentiation
6. Derivatives of Trigonometric Functions
 - a. Derivatives of Inverse Trigonometric Functions
7. Derivatives of Exponential and Logarithmic Functions

#Derivative and Differentiation

Derivative is the function that gives the instantaneous rate of change of 'y' with respect to (w.r.t.) 'x'.

Differentiation is the process of finding the derivative of a function.

Notations of Derivative: $\frac{dy}{dx}$, y' , $f'(x)$, $\frac{d f(x)}{dx}$

(Definition/First Principle) of Derivative

If $f(x)$ is a continuous function then the (derivative) or (differential coefficient) of y w.r.t. 'x' is

$\frac{dy}{dx} = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$ where Δx and Δy are the small increment in 'x' and 'y' respectively.

#Rules of Differentiation

1. **Power Rule:** $\frac{d}{dx}(x^n) = nx^{n-1}$
2. **Constant Rule:** $\frac{d}{dx}(k) = 0$
3. **Sum Rule:** $\frac{d}{dx}[f(x) \pm g(x)] = \frac{d}{dx} f(x) \pm \frac{d}{dx} g(x)$
4. **Product Rule:** $\frac{d}{dx}[f(x) g(x)] = g(x) \frac{d}{dx} f(x) + f(x) \frac{d}{dx} g(x)$
5. **Constant Coefficient Rule:** $\frac{d}{dx}[k f(x)] = k \frac{d}{dx} f(x)$
6. **Quotient Rule:** $\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{g(x) \frac{d}{dx} f(x) - f(x) \frac{d}{dx} g(x)}{[g(x)]^2}$
7. **General Power Rule:** $\frac{d}{dx}[(ax + b)^n] = na(ax + b)^{n-1}$
8. **(Chain Rule) or (Composite Function Rule):** $\frac{d f(g(x))}{dx} = \frac{d f(g(x))}{d g(x)} \times \frac{d g(x)}{dx}$

#Parametric Differentiation

The functions which are expressed in terms of a common parameter (variable) are called parametric functions.

If $y = f(t)$ and $x = g(t)$ are the parametric functions then $\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx}$

#Implicit Differentiation

Explicit Function: The function in which y can be expressed purely in terms of x called explicit function. Such function is denoted by $f(x)$.

Implicit Function: The function in which y cannot be expressed purely in terms of x is called implicit function. Such function is denoted by $f(x, y)$.

If $f(x, y) = g(x, y)$ then $\frac{d}{dx} f(x, y) = \frac{d}{dx} g(x, y)$

#Successive Differentiation

If $y = f(x)$ then

First derivative of $f(x)$ w.r.t. ' x ' is $f'(x)$ or $\frac{dy}{dx} = \frac{d}{dx}(y)$

Second derivative of $f(x)$ w.r.t. ' x ' is $f''(x)$ or $\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{d}{dx}(y)\right)$

...

n^{th} derivative of $f(x)$ w.r.t. ' x ' is $f^{(n)}(x)$ or $\frac{d^ny}{dx^n} = \underbrace{\frac{d}{dx}\left(\frac{d}{dx}\left(\frac{d}{dx}\left(\dots \frac{d}{dx}(y)\right)\right)\right)}_{n \text{ times}}$

#Derivatives of Trigonometric Functions

1. $\frac{d}{dx}(\sin x) = \cos x$
2. $\frac{d}{dx}(\cos x) = -\sin x$
3. $\frac{d}{dx}(\tan x) = \sec^2 x$
4. $\frac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$
5. $\frac{d}{dx}(\sec x) = \sec x \tan x$
6. $\frac{d}{dx}(\operatorname{cosec} x) = -\operatorname{cosec} x \cot x$

Derivatives of Inverse Trigonometric Functions

$$1. \frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$$

$$2. \frac{d}{dx}(\cos^{-1} x) = \frac{-1}{\sqrt{1-x^2}}$$

$$3. \frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$$

$$4. \frac{d}{dx}(\cot^{-1} x) = \frac{-1}{1+x^2}$$

$$5. \frac{d}{dx}(\sec^{-1} x) = \frac{1}{x\sqrt{x^2-1}}$$

$$6. \frac{d}{dx}(\operatorname{cosec}^{-1} x) = \frac{-1}{x\sqrt{x^2-1}}$$

#Derivatives of Exponential and Logarithmic Functions

$$1. \frac{d}{dx}(e^x) = e^x$$

$$2. \frac{d}{dx}(a^x) = a^x \log a$$

$$3. \frac{d}{dx}(\log x) = \frac{1}{x}$$

$$4. \frac{d}{dx}(\log_a x) = \frac{1}{x \log a}$$